BACSE202 - DATABASE SYSTEMS

Project Details

on

Receipt Aware Dual State Database Auditor

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Project Description

The project checks whether stored financial records still agree with evidence held outside the business database. The working laboratory stores each accepted change as a witness receipt containing the old state, new state and record revision. SHA-256 hashes of these receipts are batched into Merkle checkpoints. The verifier compares the live row, database audit entry, witness history and proof. A browser dashboard demonstrates legitimate updates, pending checkpoints, unauthorised edits, deleted records and rewritten audit entries. The tested version uses SQLite; the repository also retains a separate PostgreSQL and local-blockchain pipeline.

Novelty of the Idea

The proposed contribution is a reconciliation rule that treats an unanchored legitimate update differently from a database change that disagrees with a witness receipt. Before accepting another write or creating a checkpoint, the system checks revision continuity and agreement between the receipt, audit entry and current row. This prevents the demonstrated row-and-audit rewrite from being accepted simply because a new root can be computed. Tests also cover missing audit evidence, rollback and witnessed deletion. The novelty claim concerns this specific decision workflow, not SHA-256, Merkle trees or asynchronous anchoring. Close prior art exists in US20210377037A1 and US10261711B1; patent novelty and non-obviousness remain to be assessed through the institutional search.

Proposed Outcome: Patent, subject to prior-art assessment and institutional approval.

Prior art: US20210377037A1, Microsoft Technology Licensing, 2021; US10261711B1, 2019. Full references and test evidence are included in the project report.